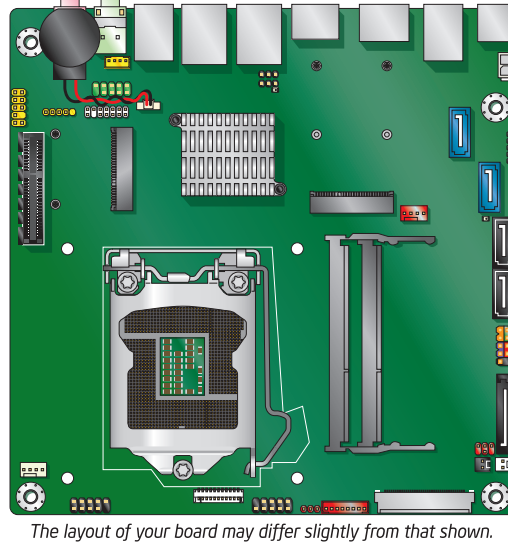


Intel® Desktop Board DQ77KB Integration Guide

This guide contains basic instructions for installing the desktop board in a compatible chassis. For a complete description of the board and its features, refer to the Technical Product Specification at: <http://www.intel.com/products/motherboard>.



The layout of your board may differ slightly from that shown.

2 Install a Heatsink

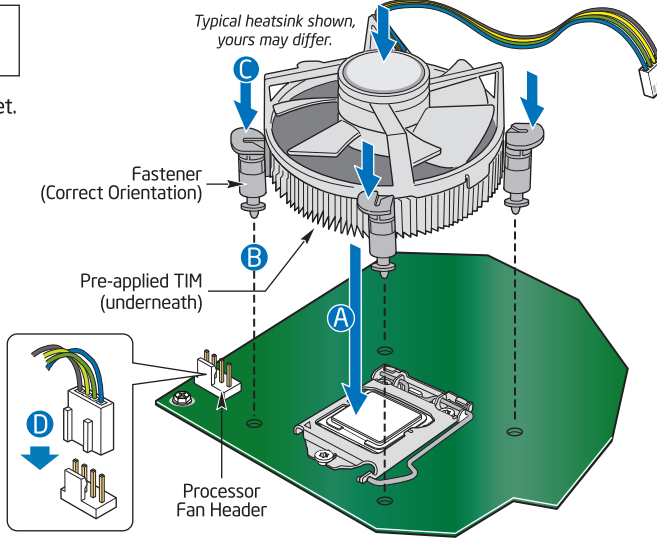
This step shows how to install both a standard height heatsink and a low-profile heatsink, choose the installation that matches your configuration.

Standard Height Heatsink

NOTE: Heatsinks that come with boxed Intel® processors use pre-applied thermal interface material (TIM) and do not need thermal grease. If you install a different heatsink, refer to the manufacturer's instructions.

Do not touch or disturb the TIM on the heatsink during installation.

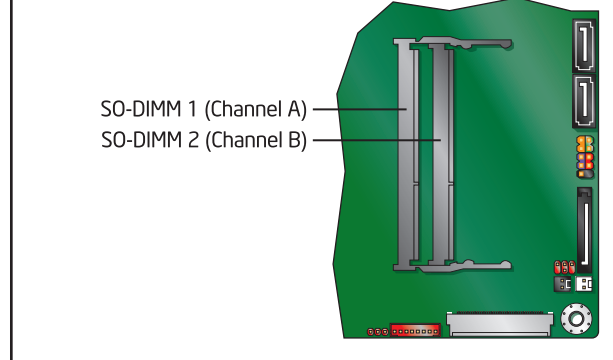
- A** Place the heatsink onto the processor socket. Ensure that the fan power cable is on the side closest to the processor fan header.
- B** Align the four fasteners with the corresponding board holes. Ensure that the fastener slots are pointing perpendicular to the heatsink.
- C** While pressing down on the heatsink, press down on the top of the fasteners with your thumb to lock into place. Ensure that all four fasteners are secured.
- D** Connect the heatsink fan power cable to the processor fan header.



5 Install System Memory

Suggested Memory Configurations and Population Order

NOTE: This desktop board supports 204-pin DDR3 SO-DIMMs only. For a list of tested memory go to: <http://www.intel.com/support/go/buildit>.



Minimum memory: 1 GB 1066 MHz DDR3 SO-DIMM. Memory should be installed in DIMM number order:

- For single-channel operation, populate Slot 1.
- For dual-channel operation, populate Slot 1 and Slot 2.

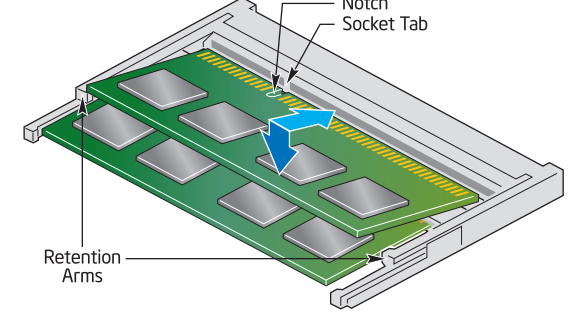
For best performance, DIMM pairs should be identical in size, speed, and organization.

Do not touch the gold contacts when handling or installing DIMMs.

NOTE: If you are installing only one SO-DIMM, it must be installed in the SO-DIMM 1 socket.

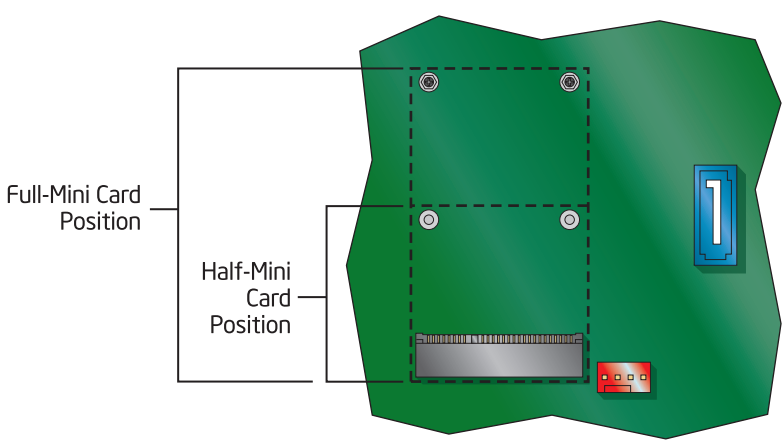
Hold the SO-DIMM with the back edge tilted slightly upwards, insert it into the socket, aligning the notch in the SO-DIMM with the tab in the socket. Gently push the back edge of the SO-DIMM down until it snaps into the retention arms.

If you are installing a second SO-DIMM, repeat this procedure.



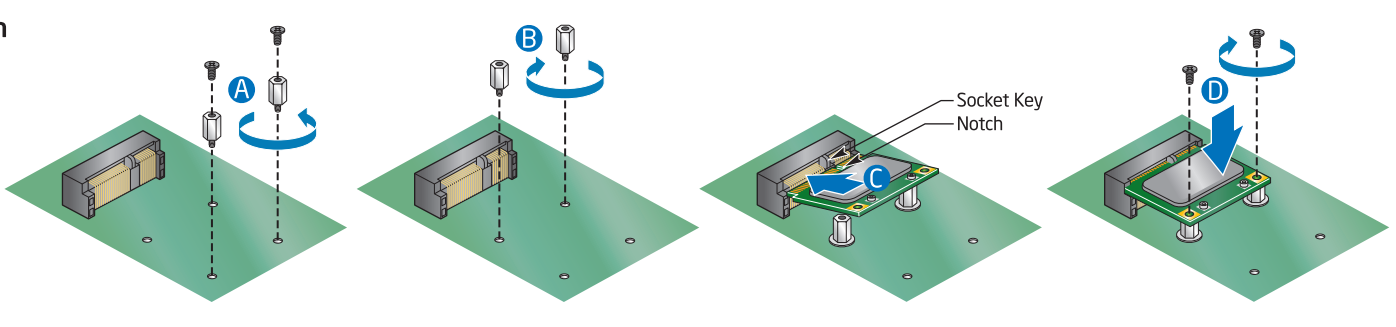
8 Install a PCI Express* Mini Card In the Full-Mini Card Slot (Optional)

This step shows how to install both a PCI Express Half-Mini Card and a PCI Express Full-Mini Card, choose the installation that matches your configuration.



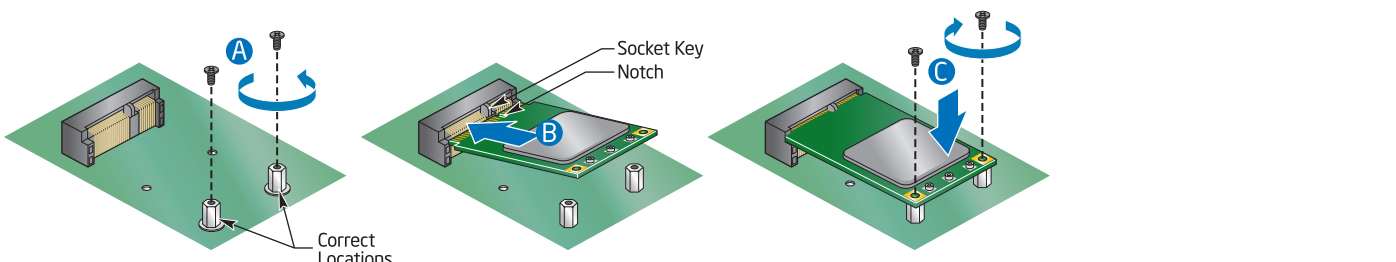
PCI Express Half-Mini Card Installation

- A** Remove the two screws and the two standoffs from the Full-Mini Card position.
- B** Install the two standoffs into the Half-Mini Card position.
- C** Align the notch in the card with the socket key and insert the card at a slightly upward angle as shown.
- D** Push down on the card and secure with two screws.



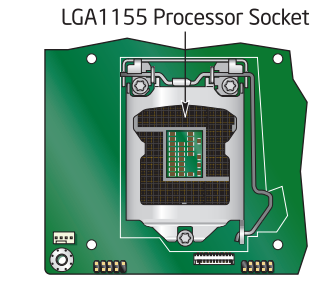
PCI Express Full-Mini Card Installation

- A** Verify that the two standoffs are in the correct locations and remove the two screws.
- B** Align the notch in the card with the socket key and insert the card at a slightly upward angle as shown.
- C** Push down on the card and secure with two screws.



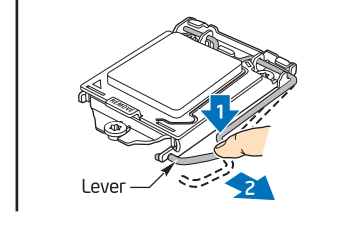
1 Install a Processor

For a list of processors this board supports, go to: <http://processormatch.intel.com>. LGA1155 Processor Socket



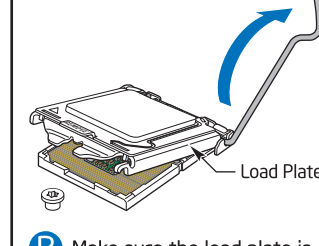
A. Unlatch the Socket Lever

Push the lever down and away from the socket to release it.

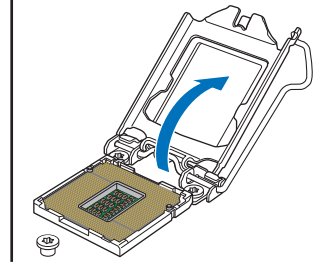


B. Open the Load Plate

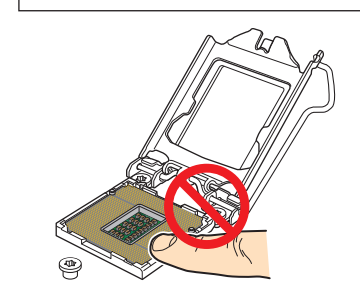
A Rotate the socket lever to lift the load plate away from the socket.



B Make sure the load plate is in the fully open position.



When opening the socket, DO NOT TOUCH the gold socket contacts.



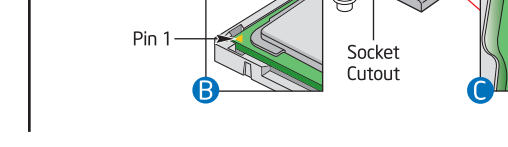
When unpacking a processor, hold by the edges only to avoid touching the gold contacts.



C. Install the Processor

The processor must align correctly with the socket before installation. DO NOT DROP the processor into the socket.

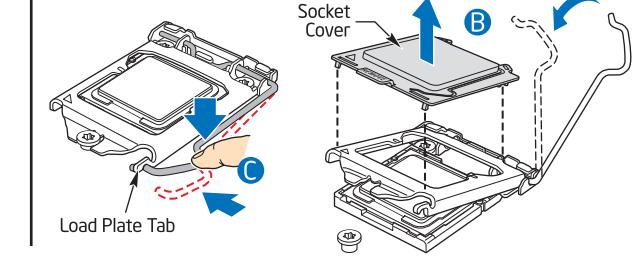
- A** Hold the processor with your thumb and index finger as shown to align your fingers with the socket cutouts.
- B** Make sure that the processor Pin 1 indicator (gold triangle) is aligned with the Pin 1 chamfer on the socket.
- C** Make sure that the notches on the processor align with the posts on the socket.
- D** Lower the processor straight down without tilting or sliding it in the socket.



D. Close the Load Plate and Secure the Socket Lever

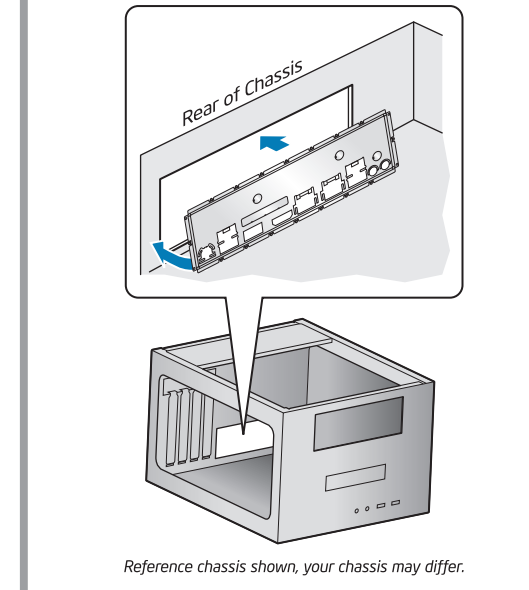
- A** Carefully lower the load plate and make sure it slides under the shoulder screw cap as the lever is lowered.
- B** Continue to lower the lever and the socket cover will pop off.
- C** Latch the socket lever under the load plate tab.
- D** Pick up the socket cover and remove it from the board.

NOTE: Save the socket cover and replace it if the processor is removed.



3 Install the I/O Shield

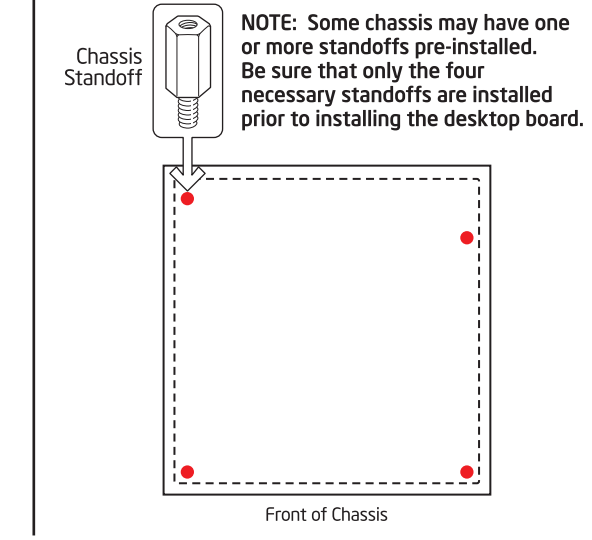
Place the I/O shield inside the chassis and press it into place so that it fits tightly and securely. Use caution so you do not deform the I/O shield.



4 Install the Desktop Board

A. Install Standoffs

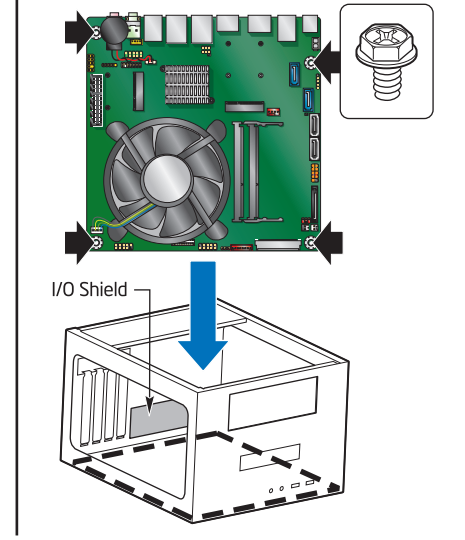
Four standoffs should be installed into the chassis before installing the desktop board. Locate the threaded standoff holes that match the desktop board, and install a standoff at each location indicated by the RED circles.



NOTE: Some chassis may have one or more standoffs pre-installed. Be sure that only the four necessary standoffs are installed prior to installing the desktop board.

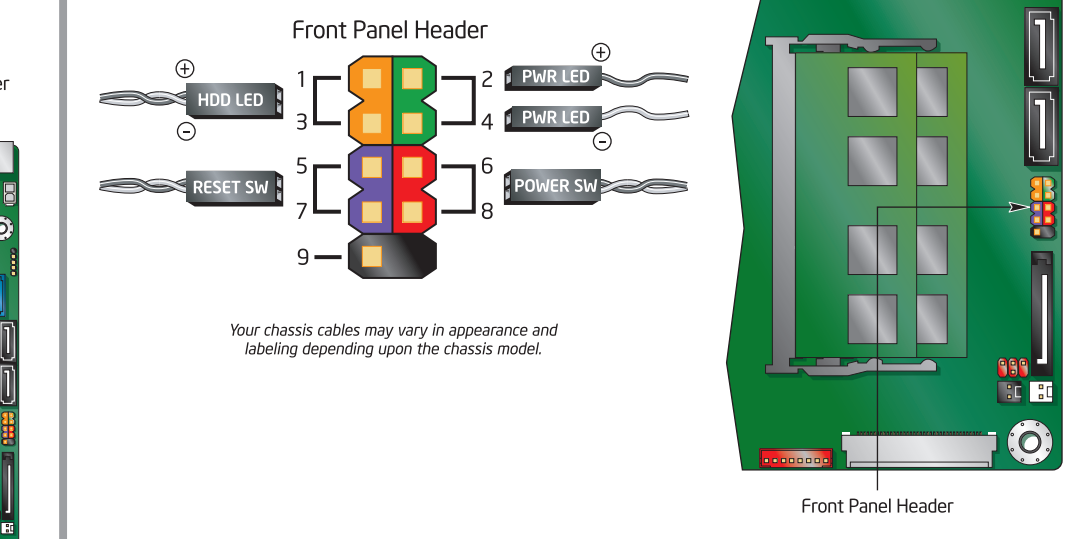
B. Install the Desktop Board

Install the desktop board by aligning the back panel with the I/O shield and securing the board to the standoffs using the screws provided with your chassis.



7 Connect Chassis Front Panel Cables

Make the front panel connections as shown in the diagram below.

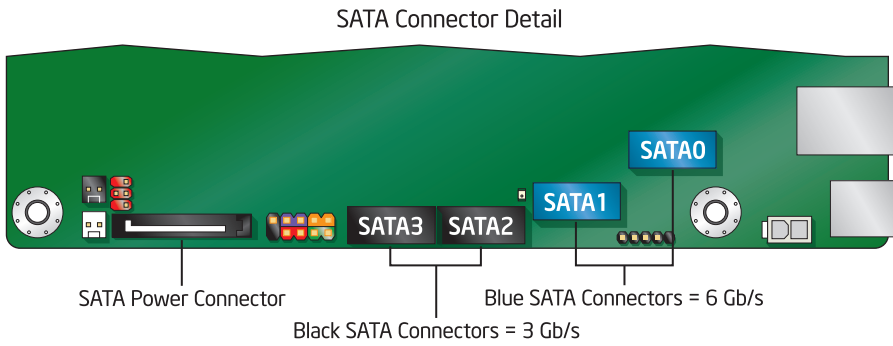


Your chassis cables may vary in appearance and labeling depending upon the chassis model.

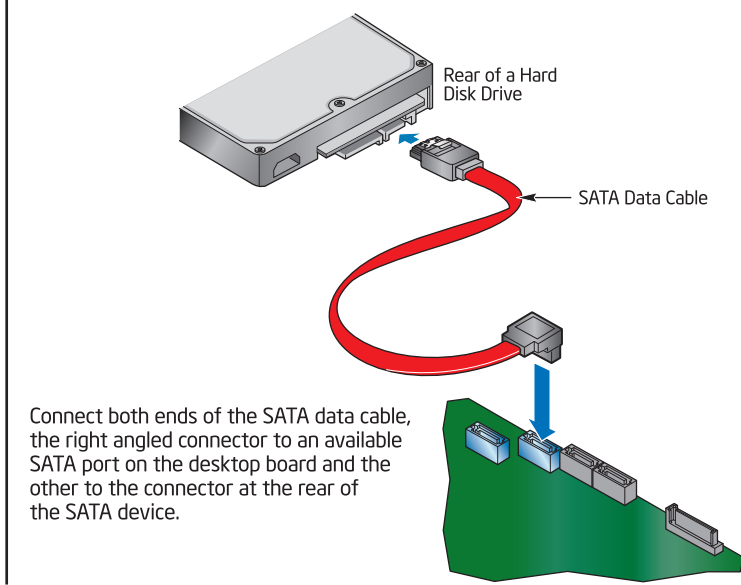
9 Install and Connect SATA Devices

A. Install SATA Devices

Install your SATA devices (Hard Disk Drive, Optical Drive, etc.). See the documentation that came with your chassis or SATA device for device installation.



B. Connect the SATA Data Cable

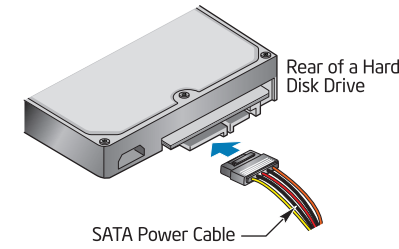


Connect both ends of the SATA data cable, the right angled connector to an available SATA port on the desktop board and the other to the connector at the rear of the SATA device.

C. Connect your SATA Device to a Power Source

Power Supply

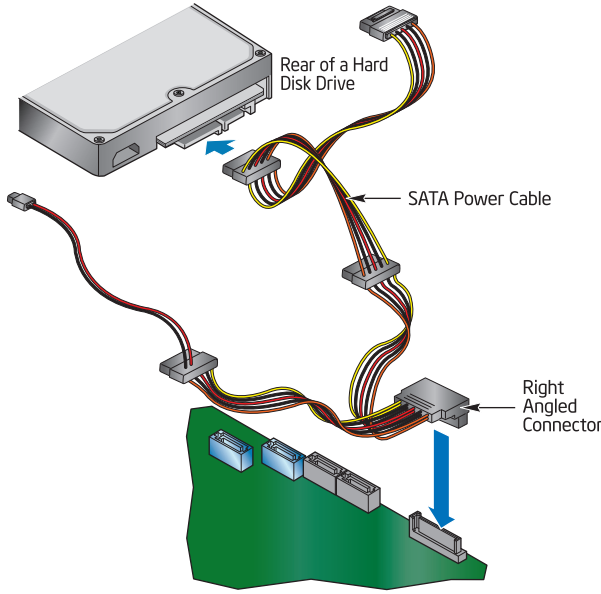
Connect the SATA power cable from the power supply to the mating connector at the rear of the SATA device.



Desktop Board Power

Using the included SATA power cable, connect the right angled connector to the SATA power connector on the desktop board and the other end to the mating connector at the rear of the SATA device.

OR

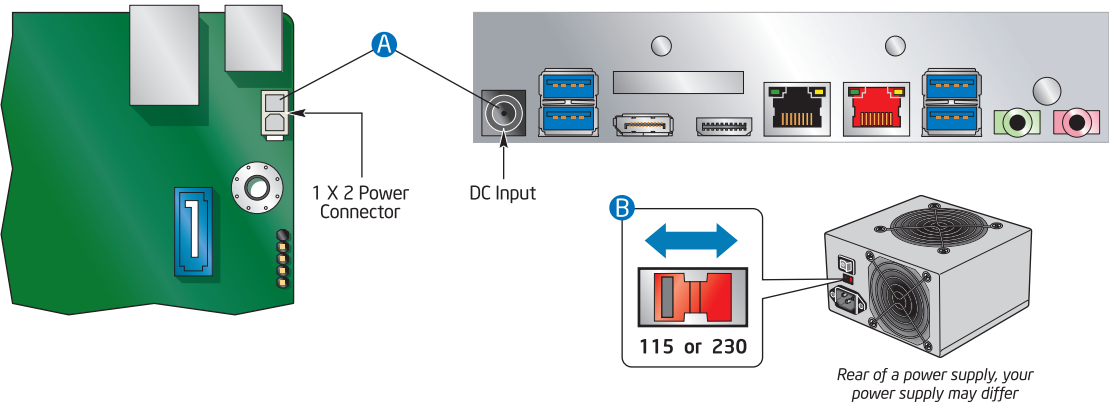


10 Make Power Connections

- A** Connect **EITHER** a 1 x 2 power supply cable to the matching 1 x 2 power connector on the board **OR** an external power supply to the DC Input connector on the back panel. **DO NOT connect both.**

- B** For a standard height chassis, ensure that the voltage setting on the rear of the power supply is set correctly.

Failure to use an appropriate power supply may result in damage to the board, or the system may not function properly.



Safety and Regulatory Information

Battery Warning

Risk of explosion if the battery is replaced with an incorrect type. Batteries should be recycled where possible. Disposal of used batteries must be in accordance with local environmental regulations.

FCC Declaration of Conformity

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

For questions related to the EMC performance of this product, contact: Intel Corporation, 5200 N.E. Elam Young Parkway, Hillsboro, OR 97124 1-800-628-8686.

Canadian Department of Communications Compliance Statement

This digital apparatus does not exceed the Class B limits for radio noise emissions from digital apparatus set out in the Radio Interference Regulations of the Canadian Department of Communications.

Le présent appareil numérique n'émet pas de bruits radioélectriques dépassant les limites applicables aux appareils numériques de la classe B prescrites dans le Règlement sur le brouillage radioélectrique édicté par le ministère des Communications du Canada.

Japan VCCI Statement

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Korea Class B Statement

이 기기는 가정용(B급) 전자파적합기기로서 주 로 가정에서 사용하는 것을 목적으로 하며, 모든 지역에서 사용할 수 있습니다.

Korea Certification mark. Includes an adjacent KCC (Korean Communications Commission) certification number: KCC-REM-CPU-DQ77KB.

Date of manufacture: March 2012
Country of Origin: China

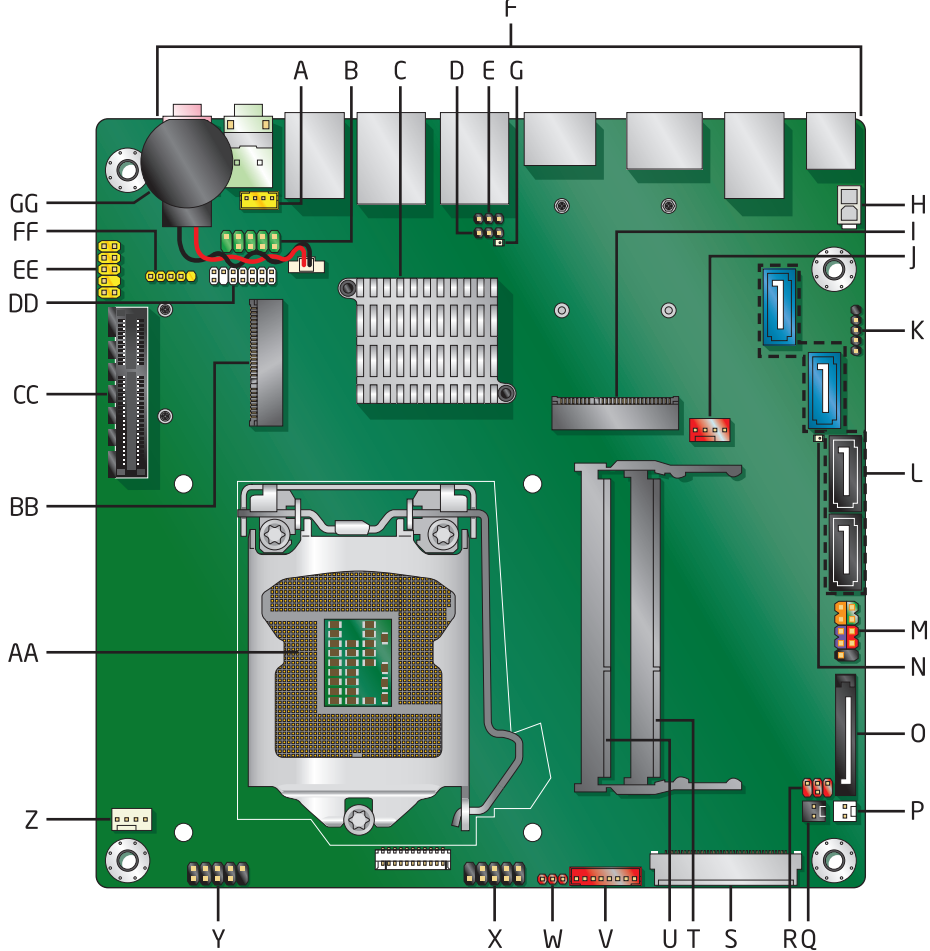
China RoHS Environmentally Friendly Use Period

The Environmentally Friendly Use Period (EFUP) for Intel Desktop Boards has been determined to be 10 years.

For detailed information about the desktop board's regulatory compliance, refer to the Technical Product Specification at: <http://www.intel.com/products/motherboard>.

Reference

Desktop Board Components



- A. Internal Stereo Speakers Connector
B. Serial Port Header
C. Intel® Q77 PCH
D. MEBX_RESET Jumper
E. BIOS Configuration Jumper
F. Back Panel Connectors
G. ME_STATE LED
H. 1 x 2 Power Connector
I. PCI Express Full-Mini/Half-Mini Card Slot
J. System Fan Header
K. USB 2.0 Front Panel Header (Single)
L. SATA Data Connectors
M. Front Panel Header
N. Standby Power LED
O. SATA Power Connector
P. PANEL_OFF Header
Q. Chassis Intrusion Header
R. Panel Voltage Selection Header
S. LVDS Connector
T. SO-DIMM 2 (Channel B)
U. SO-DIMM 1 (Channel A)
V. FPD Brightness Connector
W. Backlight Inverter Voltage Selection Header
X. USB 2.0 Front Panel Header (Dual)
Y. USB 2.0 Front Panel Header (Dual)
Z. Processor Fan Header
AA. Processor Socket
BB. PCI Express Half-Mini Card Slot
CC. PCI Express 3.0 x4 Connector
DD. Custom Solutions Header
EE. Front Panel HD Audio Connector
FF. DMIC Header
GG. Battery

Header and Connector Pinouts

Front Panel HD Audio

Port1L 1 2 Ground
Port1R 3 4 Presence#
Port2R 5 6 Sense1_Ret
Sense_Send 7 Key (no pin)
Port2L 9 10 Sense2_Ret

Front Panel AC '97 Audio

MIC 1 2 AUD_GND
MIC_BIAS 3 4 PRESENCE#
FP_OUT_R 5 6 AUD_GND
No Connection 7 Key (no pin)
FP_OUT_L 9 10 AUD_GND

Internal Stereo Speakers

1 Front_L-
2 Front_L+
3 Front_R+
4 Front_R-

DMIC

+3.3 V 1
DMIC_DATA 2
Ground 3
DMIC_CLK 4
Key (no pin)

USB 2.0 Front Panel (Dual)

Power (+5 V) 1 2 Power (+5 V)
D- 3 4 D-
D+ 5 6 D+
Ground 7 8 Ground
Key (no pin) 9 10 No Connection

Front Panel USB 2.0 (Single)

Key (no pin)
Ground 4
D+ 8
D- 2
Power (+5 V) 1

Serial Port

DCD 1 2 RXD#
TXD# 3 4 DTR
Ground 5 6 DSR
RTS 7 8 CTS
RI 9 Key (no pin)

Front Panel

HD LED - 1 2 + Power LED
Reset 5 6 On/Off
+5 V - 9

PANEL_OFF

Monitor_off 2
Ground 1

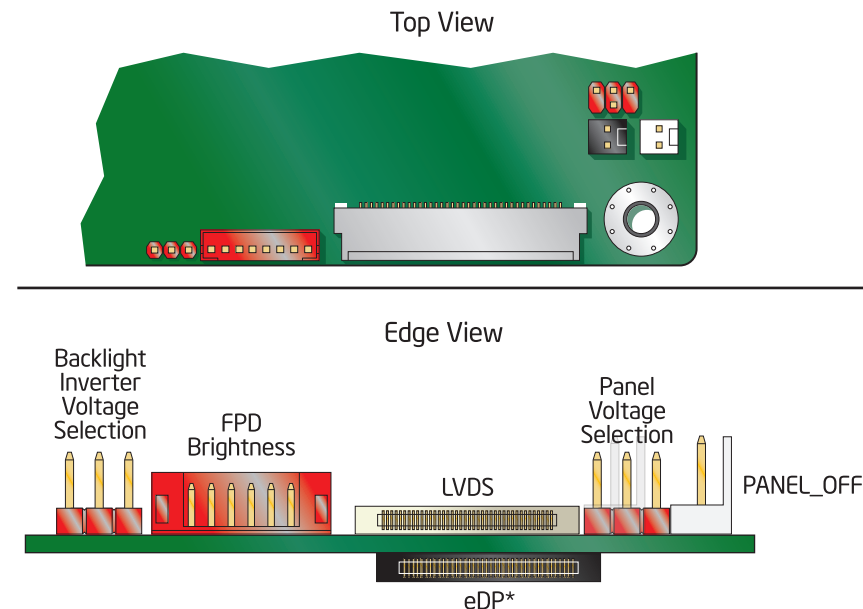
Chassis Intrusion

Intruder# 2
Ground 1

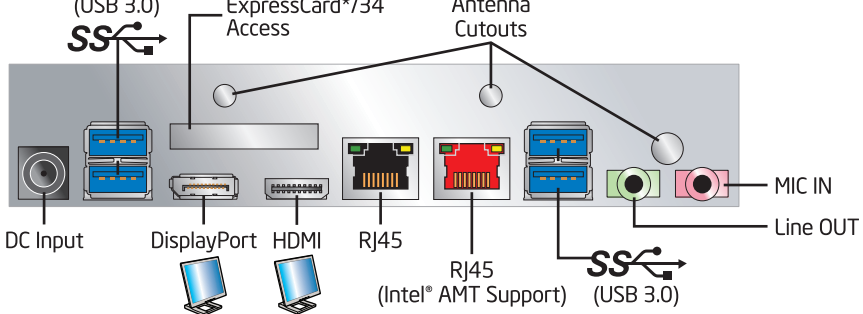
Custom Solutions

Prog_LED 1 2 Ground
Key (no pin) 4 SMB_CLK
+3.3 VSB 5 6 SMB_Data
PWRBT# 7 8 HDMI_CEC
+5 VSB 9 10 No Connection
USB+ 11 12 USB-
SCI/SMI GPIO 13 14 WDIO#/GPIO

Flat Panel Display Headers/Connectors



Back Panel Connectors



BIOS Reference

The BIOS (Basic Input/Output System) controls the computer's boot process. The purpose of the BIOS is to identify and initialize processor, memory, hard drives, optical drives, and other hardware.

BIOS Settings

For a list of BIOS settings along with their purpose and options, refer to the BIOS Glossary at: <http://www.intel.com/support/motherboards/desktop/sb/cs-020304.htm>.

Updating the BIOS

You should update the BIOS on your board only if the newer BIOS version solves a specific problem you have. BIOS updates are available in Intel's Download Center at: <http://downloadcenter.intel.com/>.

There are various methods of updating an Intel® Desktop Board BIOS to the latest version. The number of methods available for any particular board model varies, depending on drive support and BIOS update file size. For update instructions, go to: <http://www.intel.com/support/motherboards/desktop/sb/cs-022312.htm>.

Troubleshooting the BIOS

For tips on troubleshooting BIOS issues on Intel® Desktop Boards, refer to: <http://www.intel.com/support/motherboards/desktop/sb/cs-028780.htm>.

BIOS Configuration Jumper Settings:

1-2	Normal
2-3	Configuration
No jumper	Recovery

MEBX_RESET Jumper Settings:

1-2	Reset MEBX
2-3	N/A
No jumper	Normal

Online Support

For more information on Intel Desktop Board DQ77KB, consult the following online resources:

General board information <http://www.intel.com/products/motherboard/index.htm>
Available board configurations <http://ark.intel.com>
Supported processors <http://processormatch.intel.com>
Chipset information <http://www.intel.com/products/desktop/chipsets/index.htm>
BIOS and driver updates <http://downloadcenter.intel.com/>
More integration information <http://www.intel.com/support/go/buildit>
Customer support http://www.intel.com/p/en_US/support?iid=hdr+support
Intel® Rapid Storage Technology http://www.intel.com/p/en_US/support/highlights/chpsts/ismm
Tested memory <http://www.intel.com/support/motherboards/desktop/sb/cs-025414.htm>

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